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Machine Diagnostic System and Method for Vibration Analysis on Radial Drilling Machine

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Abstract

Machine vibration is essentially the forward and backward development of machines or machine segments. Any segment that moves forward and backward or wavers is vibrating. Machine vibration can capture in different structures. All the more regularly, vibration is bothersome, squandering vitality and making the undesirable sound. Vibration investigation is an instrument to recognize, avert and anticipate disappointments in pivoting apparatus, for example, spiral boring machine, machine, and processing machine and so on. Actualizing vibration examination on these machines will enhance the unwavering quality of the machines and lead to all the more likely machine proficiency and diminished downtime disposing of mechanical or electrical part disappointments. Regular machines incorporate engines, siphons, fans, outfit boxes, blowers, turbines, transports, rollers, motors and mechanical devices having rotational components. The turning components of these machines produce vibrations at explicit frequencies that recognize the pivoting components capacities. In the present work, it is proposed to break down the vibration on spiral boring machine utilizing a piezoelectric sensor. Piezoelectric sensors are case mounted utilizing a perpetual jolt, convenient magnet or cement to hold them set up. They will gauge the vibration of the machine and deliver yield as a voltage or current corresponding to the vibration.

Keywords: Pumps, Compressors, Conveyors, Piezoelectric, Accelerometers, Displacement, Probes RDM.

1. Introduction

In the examination of the dynamic behavior of machine apparatuses rigid nature' and 'Strength' of machine device ought to be considered. Dynamic Rigidity is characterized as the proportion of amplitude of vibratory power to vibratory removal at a given recurrence. Dynamic unbending nature is the proportion of structures protection from vibrations and it alludes to consistent state vibrations. Dynamic security relates to transient reaction qualities. Dynamic solidness demonstrates that the majority of a disseminated framework wavers with diminishing amplitudes towards the first condition of balance and establishes a dissipative framework. Consequently, it is important to contemplate the wellsprings of vibrations and their belongings. This investigation will supply the data about the greatest decent dimension of Vibrations and after that machine, the device can be structured so that wellsprings of vibrations can be wiped out or its power can be diminished. In the present examination, a few parts of dynamic conduct of Radial Drilling Machine have been considered through tests and investigation of trial results have been done in various approaches to giving some helpful information

2. Literature Review

A large number of investigations have been undertaken by

S.A. Tobias [1] and J. Tlutsy [2] to predict the maximum chatter free width of cut. The accurate prediction of this parameter depends on the dynamic cutting forces, which vary with the cutting conditions. S.A. Tobias and W. Fishwick [3], proposed the first analytical expression for dynamic cutting force considering the incremental variation of chip-thickness, thus the rate of penetration and cutting speed. G.S. Kainth [4] have proposed another analytical model of dynamic forces considering the characteristics of the friction on the tool rake face and the shear angle oscillations in dynamic cutting. Jeon Suk Kim and Byung Ho Lee [5] presented an analytical model of dynamic forces in orthogonal cutting by expressing the same in terms of static cutting co-efficient which depends on cutting variables. The author

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verified the proposed model and found good agreement between the stability limits predicted by the theory and the critical widths of cut obtained from experiments. The investigations on the chatter stability, its reasoning and vibration suppression had been made by many researchers such as Nejat Olgae and Martin Hosek [6], S. Doi and S. Kato [7], Y. Altintas and E. Budak [8]. And a new spindle speed regulation approach to online control of self-excited regenerative chatter vibration during machining process is proposed by Y.S.Liao and Y.C. Young [9]. Y. Kondo, O. Kawano and H. Sato [10] pointed out that the behavior of the self-excited vibration was governed by the multiple regenerative effects. S.A. Tobias and S. Taylor [11] predicted vibration characteristics of machine tool structure with the help of lumped parameter flexibility model. N. Kasahara [12] studied the behavior of the self-excited chatter from the 'Space Phase' point of view the space phase is defined as that between outer and inner modulation, identifiable by the trace of the vibration left on the work. E. Marui and others [13] have studied the effects of vibratory characteristics and cutting conditions on the initiation of primary or regenerative chatter vibrations. And they have experimentally examined the mechanism by which chatter vibrations occur. Chatter vibrations of long drills have been investigated by S. Ema [14], using drills with different overhang lengths and special drills with different pieces of additional mass. The experimental results showed that the vibration is a regenerative chatter and its frequency is equal to the bending natural frequency of the drill when the drill point is supported in a machined hole. Based upon the experimental results, the stability of chatter vibration was discussed. It was clarified that unstable range appears only when the chatter frequency in one work-piece rotation is an odd number. H. Fujii and others [15 and 16] investigated that one of the most common operations for boring holes in metals in the drilling process, long drills, tend to have whirling vibrations and chatter vibrations because their bending rigidity and viscosity damping characteristics are extremely low. Galloway [17] found that the torque and thrust force acting on drill point in the final stage of drilling cause an axial displacement of the drill. Tobias [18] reported that a certain type of chatter vibration, which causes a relative displacement of the saddle in relation to the machine table, might be observed only in a certain speed range.

3. Experimental Setup

In the present experimental setup, an accelerometer has been attached to the drilling machine, to measure the vibration signal which is then transferred to the Arduino that converts the analog signals to the digital signals which are further transmitted and stored in a computer connected through USB port.

During the experiment, the accelerometer has been used for detecting vibration from radial drilling machine. The accelerometer detects machine vibrations of even smaller intensities and it can measure vibrations in three directions.

After that, the Arduino board is connected to the system through the USB port, on which data is collected.

Apart from this setup, an oscilloscope is also attached, just to justify whether the data collected is true or not.

Experimental Procedure:

1. Select a suitable method to connect accelerometer with the machine. For this, as shown in figure 4.4 a point P is selected where the observations are taken
2. Position the accelerometer for three mutually perpendicular directions X, Y and Z.
3. Connecting the pins of Accelerometer with Arduino board connecting, CS with VCC with help of 10-kilo ohm resistance, SCL point with pin A5, connect SDA with A4, VCC with 3.4V pin finally connect GND with the ground.
4. Interface accelerometer with Arduino board.
5. Connect Arduino with the system with help of data cable.
6. Calibrate the accelerometer with an oscilloscope and find any variation, manipulate the error and obtain the reading.



Figure 1 Experimental setup using the accelerometer

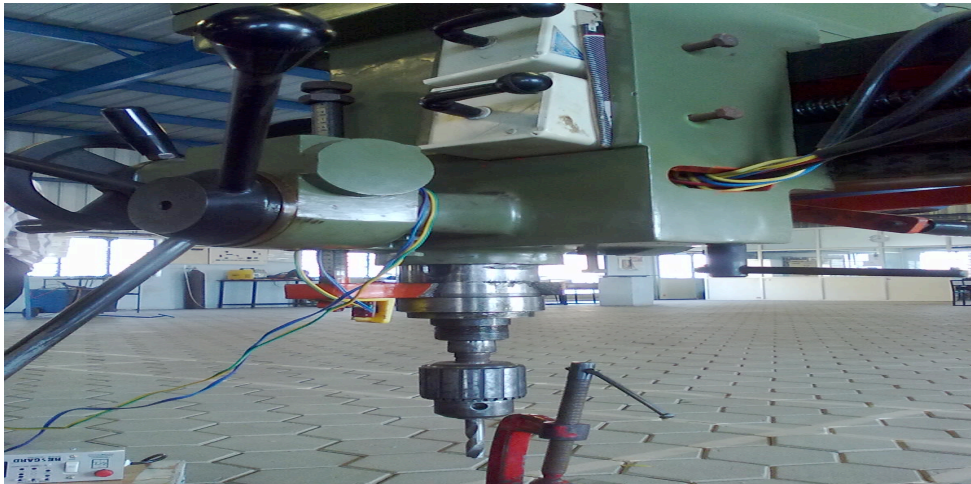


Figure 2 position of the accelerometer



Figure 3 Calibration using the oscilloscope

4. Findings

The present research contains all the observations or the numerical data which have been collected during this project work. In this, observations at 660 rpm were included for aluminum material.

Observations for Existing Spindle Bearing:

OBSERVED VALUES OF VIBRATION SIGNAL(peak to peak) at 660 RPM				
S.N.	FEED mm/Rev	Vibration in X-Direction (mV)	Vibration in Y-Direction (mV)	Vibration in Z-Direction (mV)
1	0.0198	30.2	16	52.4
2	0.0396	27.3	15.9	48.8
3	0.0594	28.6	16.6	51.6
4	0.0792	28	17.3	53.2
5	0.099	29.9	18.6	52
6	0.1188	30.3	18.9	55.1
7	0.138	31.6	20.2	56.4
8	0.158	31.9	22	57.3

Table 1 Observations at 660 rpm

Observations for New Spindle Bearing:

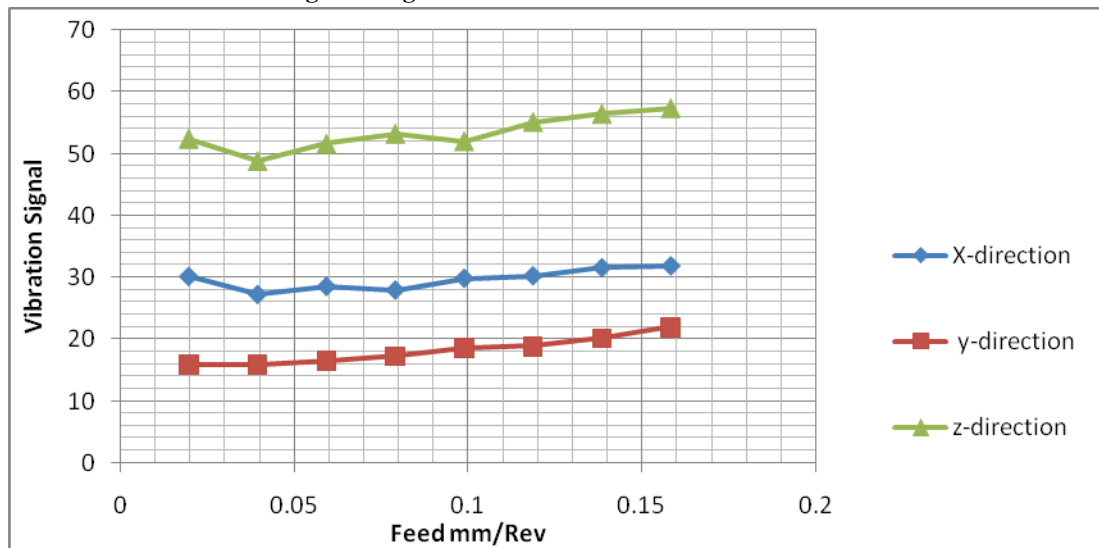
EXPERIMENTAL VALUES OF VIBRATION SIGNAL(peak to peak) at 660 RPM ON NEW SPINDLE BEARING				
S.N.	FEED mm/Rev	Vibration in X-Direction (mV)	Vibration in Y-Direction (mV)	Vibration in Z-Direction (mV)
1	0.0198	25.2	12.1	40.3
2	0.0396	24.5	12.4	39
3	0.0594	25.4	12.9	42.2
4	0.0792	26.7	14.1	44.2
5	0.099	27	15.6	45.1
6	0.1188	27.8	16.4	44.6
7	0.138	27.5	18.4	47.3
8	0.158	29.5	19.1	49.2

Table 2 Observations at 660 rpm

5. Result

The results of the sets of experiments described above have been plotted in the form of Line Graphs, Column Charts

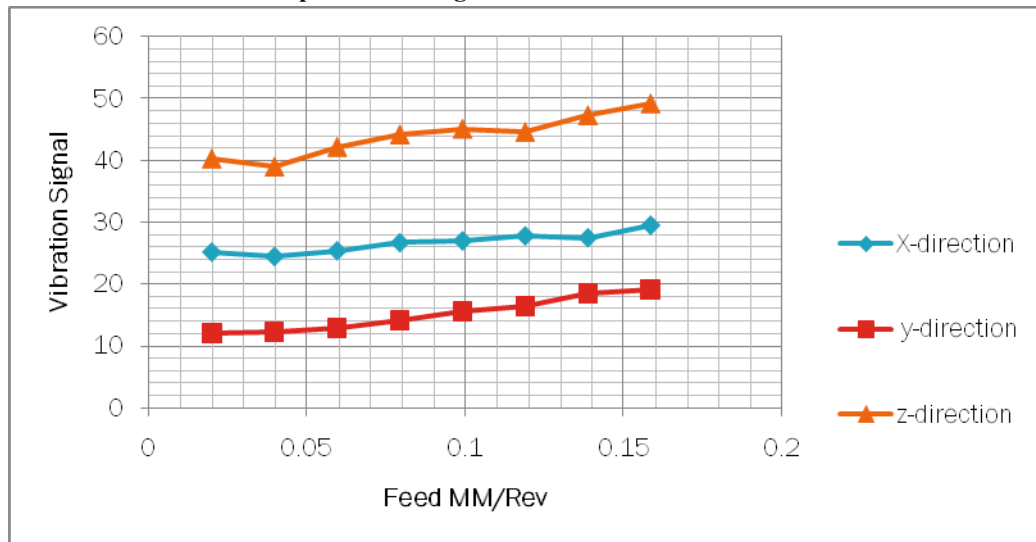
The result of Vibration with Existing Bearing:



Graph 1 Relation between vibration and feed at 660 rpm

Vibration in X, Y and Z direction increases with an increase in feed rate. Also, Amplitude of vibration is maximum (48.8 to 57.3) in the Z direction and minimum (15.9 to 22) in the Y direction at a constant speed

The result of Vibration with New Spindle Bearing:

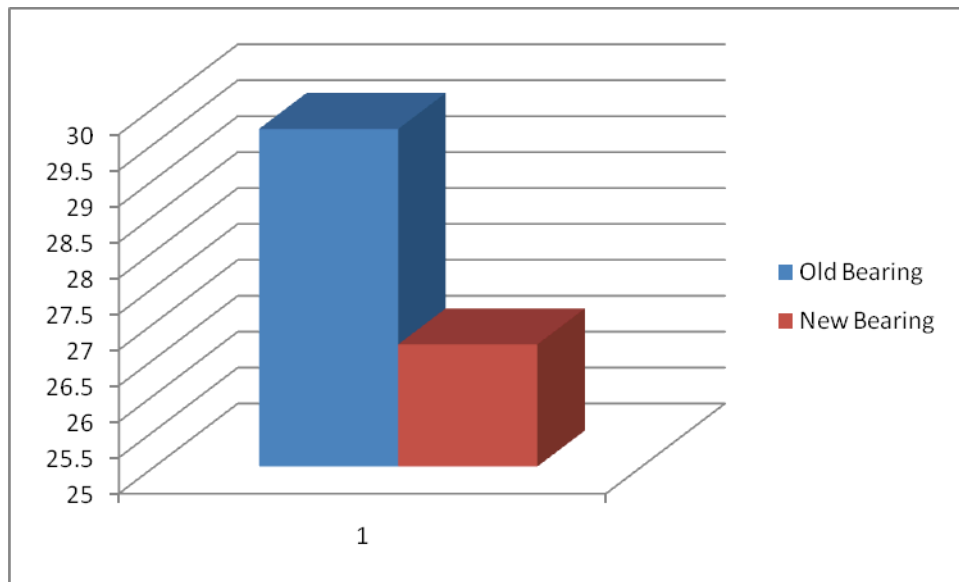
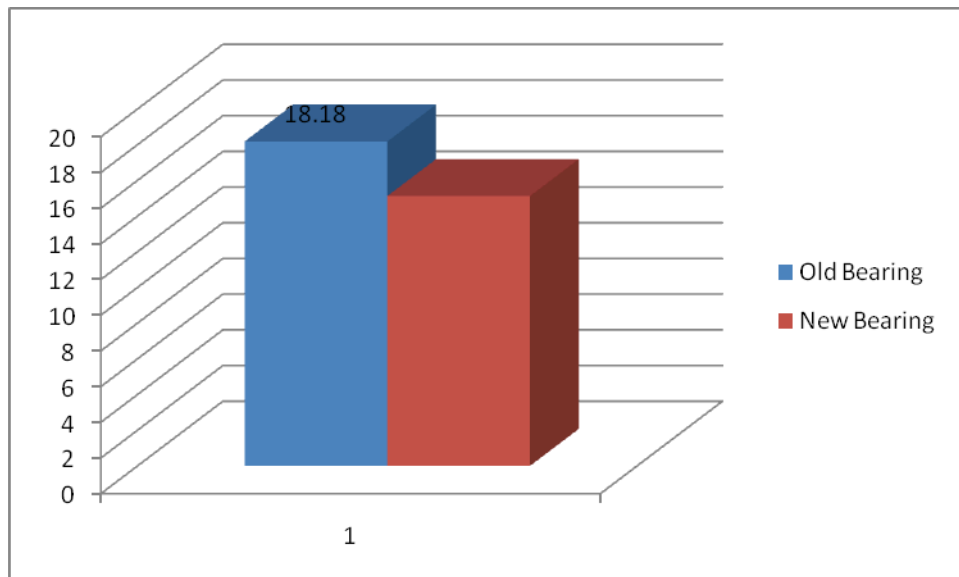


Graph 2 Relation b/w Vibration and Feed at 660 RPM

Vibration in X, Y and Z direction increases with an increase in feed rate. Also, Amplitude of vibration is maximum (39 to 49.2) in the Z direction and minimum (12.1 to 19.1) in the Y direction at constant speed.

Result table for % reduction in vibration after replacement of bearing

Speed→ Direction ↓	660 rpm		
	Old	New	% Reduction
X	29.8	26.7	10
Y	18.1	15.1	16.4
Z	53.3	43.9	17.5

Table 3 Result table**Representation of Vibration with Existing and New Bearing****Figure 4 Representation of Vibration in X-direction with Existing and New Bearing:****Figure 5 Representation of Vibration in Y-direction with Existing and New Bearing:**

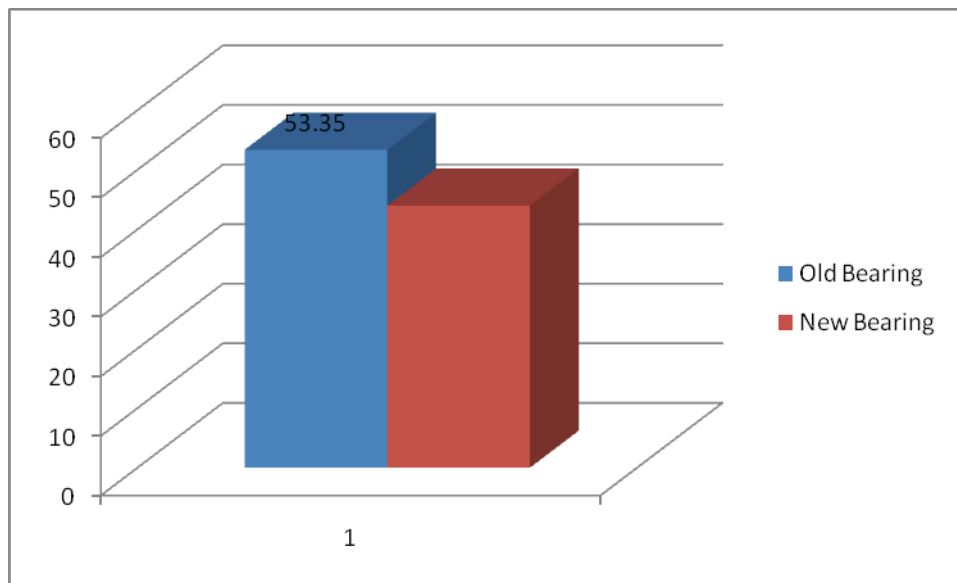


Figure 6 Representation of Vibration in Z-direction with Existing and New Bearing:

6. Discussion

Effect of Feed:

The results relating to the relationship between feed and vibration in the directions X, Y and Z at the location P, near the spindle, are shown graphically in the above figures. The set of data in these figures corresponds to the conditions of drilling with rpm of 660 and constant drill diameter of 12 mm. The figure indicates that, with an increase in feed, vibration signals in terms of mV have generally increased, though not very consistently.

The relationship between the vibrations and feed, in all the three direction X, Y and Z, has been shown at rpm 660. The 3D column charts show the comparison between the observed values of vibration signals with existing spindle bearing and new spindle bearing individually in X, Y and Z directions. It can be seen from the figures that the vibrations are somewhat more in the Z direction as compared to the X direction and Y direction.

Effect of Rotational Speed of Spindle (rpm)

The above figures illustrate the observed data in the form of spindle rpm versus vibration signals at location P in the different direction of measurement. It is seen from the figures that the increase in rotational speed of spindle increases the level of vibrations measured in all the three direction X, Y, and Z.

The figures show the relationship equations between the RPM and vibration signals observed values. These values are compared in the form of column chart in the above figures for directions X, Y and Z respectively. It can be seen from these figures that like the increase of spindle speed increase the vibration signal.

Table 3 shows the vibration for the RPM values 660. The values of % decrease of vibrations in the X, Y and Z direction.

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